INFO 7375 · PROMPT ENGINEERING FOR GENERATIVE AI · SPRING 2026

SAGE: Secure AI Governance Engine

Enterprise Compliance Policy Reasoning Agent

Assignment 3 - System Prompt Setup

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Model: GPT-4o (OpenAI API) · temperature = 0.3 · temperature = 0.5 (Self-Consistency runs)

Abstract

This report documents the systematic engineering of SAGE's (Secure AI Governance Engine) system prompt through the application of twelve prompting techniques: seven foundational and five advanced, evaluated against a single consistent test query on TechNova Inc.'s corporate policy corpus (1,602 words across three documents). Using GPT-4o via the OpenAI API, each technique was executed programmatically in Google Colab and evaluated on policy coverage completeness, citation accuracy, risk classification correctness, and response structure fidelity. The progression from a zero-shot baseline to a production-ready advanced system prompt is documented through actual model outputs, token economics, and principled cross-technique analysis. Key findings include: few-shot prompting transfers behavioral schema more reliably than structural instructions; chain-of-thought uniquely achieves exhaustive multi-document coverage but introduces risk definition ambiguity; self-consistency diagnoses prompt-level definitional instability; and self-criticism detects sub-hallucination precision degradation that standard groundedness checks miss. The final system prompt incorporates twelve discrete improvements, each attributable to a specific technique, producing a compliance reasoning agent that is grounded, auditable, and adversarially robust.

1. App Concept and Scope

SAGE is an enterprise compliance policy assistant for TechNova Inc., designed to answer employee and compliance officer queries grounded exclusively in three authoritative internal documents: the Remote Work Policy (POL-RW-2025), Data Privacy Policy (POL-DP-2025), and Information Security Policy (POL-IS-2025). The system addresses a documented enterprise failure mode: employees defaulting to informal guidance that misinterprets policy, creating regulatory exposure and security incidents that could have been prevented by accurate, cited policy interpretation.

Each SAGE response is structured to include a comprehensive policy-grounded answer, specific citations with section numbers, a unified risk classification (Low / Medium / High), and explicit reasoning connecting evidence to conclusion. This structure was chosen to support both compliance accuracy and organizational accountability; responses that cite their evidentiary basis can be reviewed, corrected, and audited, whereas conversational answers cannot.

The implementation uses the OpenAI GPT-4o API accessed via Python in Google Colab. This assignment documents the development of SAGE's system prompt through the systematic application of all course prompting techniques. Per instructor feedback on the project proposal, scope has been bounded to the three policy documents listed above as the MVP corpus, with the full RAG pipeline, adversarial security layers, and Promptfoo evaluation suite constituting subsequent milestones.

CONSISTENT TEST QUERY, APPLIED UNIFORMLY ACROSS ALL TWELVE TECHNIQUES

An employee wants to work remotely from Portugal for 45 days. They will be handling customer data and plan to use their personal laptop with a VPN connection. What policies apply, what approvals are needed, and what is the compliance risk level?

Policy Document	ID	Words	Key Sections Tested
Remote Work Policy	POL-RW-2025	452	§4.1, §4.2, §4.3, §4.4, §4.5, §5.2
Data Privacy Policy	POL-DP-2025	585	§5.1, §5.2, §5.3, §5.4
Information Security Policy	POL-IS-2025	565	§4.1, §4.2, §5.1, §5.3, §6.1–6.5
Total Corpus	—	**1,602**	All sections applicable to scenario

2. Initial System Prompt, Foundational Techniques

Each technique was implemented as a distinct system prompt in Python, executed against GPT-4o, and evaluated on four dimensions: (1) policy coverage completeness relative to the known applicable sections, (2) citation precision, (3) risk classification correctness, and (4) response structure fidelity. Actual model outputs are reproduced verbatim below. Token counts and response times are drawn from API response metadata.

T1 · Zero-Shot Prompting, Foundational Baseline

SYSTEM PROMPT

> You are a compliance policy assistant for TechNova Inc. Answer questions based ONLY on the following policy documents. If the policies do not contain enough information to answer, say so clearly. POLICY DOCUMENTS: [ALL_POLICIES]

Tokens: 2,721 · Response time: 5.87s · Output: 292 words · Temperature: 0.3

MODEL RESPONSE (GPT-4O)

> To work remotely from Portugal for 45 days, the following policies and approvals apply: 1. Remote Work Policy (POL-RW-2025), Section 4.2: Duration exceeds 30-day limit; approval required from direct manager plus HR and Legal. Section 4.3: Compliance with POL-DP-2025 and POL-IS-2025 is mandatory at all times. 2. Data Privacy Policy (POL-DP-2025), Section 5.3: Customer data must remain within approved systems; not stored on local devices unless encrypted. Portugal is in the EEA; cross-border transfer rules require review. 3. Information Security Policy (POL-IS-2025), Section 6.1–6.2: BYOD registration required; device must meet encryption and endpoint protection requirements. Section 4.1: All remote connections must use the company VPN. Compliance Risk Level: - Data Security Risk: Moderate (handling customer data on personal device) - Regulatory Risk: Moderate (extended international work, multi-department approvals) - Operational Risk: Moderate (approval complexity across HR, Legal, IT Security)

Analytical Evaluation: Zero-shot establishes an accurate but structurally deficient baseline. The model correctly identifies all three applicable policy documents and applies the 30-day threshold from §4.2 without prompting, demonstrating that GPT-4o's attention mechanisms reliably retrieve high-salience policy triggers in a zero-shot context. However, two structural deficiencies emerge with direct implications for production use. First, the risk assessment fragments into three domain-specific sub-risks rather than producing a unified compliance classification, undermining actionability for the non-specialist employee who receives the response. Second, §4.4 (benefits and health insurance coverage gap for international arrangements) and the §5.1 EEA adequate safeguard framework are entirely absent, not because they were retrieved and deprioritized, but because zero-shot attention is non-exhaustive across a multi-section, multi-document corpus. This reflects the fundamental limitation of single-pass holistic generation: the model attends to what the query makes salient, not to what a complete compliance review requires.

T2 · Few-Shot Prompting, Behavioral Schema Transfer

SYSTEM PROMPT (TWO LABELED EXAMPLES INCLUDED)

> You are a compliance policy assistant for TechNova Inc. [POLICY DOCUMENTS] EXAMPLE 1: User: What is the data retention period for customer PII? Assistant, Answer: Customer PII must be retained for no longer than 7 years following the end of the customer relationship, unless a longer period is required by law. Citations: - POL-DP-2025, Section 4.1, 7-year retention limit for customer PII Risk Level: Low Reasoning: Single-section lookup with no cross-policy

dependencies; requirement is unambiguous, and requirement is met by default retention practices. EXAMPLE 2: User: Can an employee install a third-party project management tool on their company laptop? Assistant, Answer: No. Prior IT Security approval is required before installing any software on company devices. Citations: - POL-IS-2025, Section 3.3, Unauthorized software installation is prohibited. Risk Level: Medium Reasoning: Procedural violation; risk is elevated by potential vulnerability introduction but remediable through the IT Security approval process. Now answer the following question using the same format:

Tokens: 2,807 · Response time: 5.26s · Output: 173 words · Best format fidelity · Correct unified risk

MODEL RESPONSE (GPT-4O)

Answer: The employee must obtain prior written approval from their direct manager for international remote work from Portugal. Since the duration is 45 days, exceeding the 30-day threshold, additional approval from both HR and the Legal department is required. Customer data must remain within approved cloud systems and must not be stored locally on the personal laptop. The personal laptop triggers BYOD registration with the IT Security team, with device-level encryption, endpoint protection, and remote wipe capability required before access. VPN use is required for all remote connections. Citations: - POL-RW-2025, §4.2, International remote work >30 days requires HR and Legal approval - POL-RW-2025, §4.3, Compliance with Data Privacy and Info Security Policies required - POL-DP-2025, §5.1, §5.3, EEA adequate safeguards and encrypted cloud-only access - POL-IS-2025, §6.1, §6.2, BYOD registration and device security requirements Risk Level: High Reasoning: Multiple simultaneous compliance obligations across all three policies, multi-department approvals, data privacy safeguards, and BYOD security requirements, with material violation risk if any single element is neglected prior to departure.

Analytical Evaluation: Few-shot is the most efficient foundational technique by output-quality-to-cost ratio. Two labeled examples were sufficient to transfer the Answer/Citations/Risk Level/Reasoning schema completely, without explicit format instructions, confirming that in-context behavioral demonstration is a more reliable output-distribution shifter than declarative structural directives for structured generation tasks. Critically, few-shot also produced the only foundational response to correctly classify risk as a unified "High" with a compound justification, suggesting the model inferred from the examples' risk logic that multi-policy violations warrant elevated classification. The mechanism is theoretically coherent: the examples function as implicit demonstrations of the risk classification decision boundary, not just format templates. Trade-off: coverage is bounded by the examples' implicit domain emphasis; the response correctly cites §5.1 but misses §4.4 (benefits gap) and §4.5 (discouragement clause), domains not represented in the examples. Few-shot optimizes format fidelity and risk accuracy; completeness requires augmentation through complementary techniques.

T3 · Chain-of-Thought (CoT) Prompting, Exhaustive Sequential Reasoning

SYSTEM PROMPT (7-STEP MANDATORY REASONING SCAFFOLD)

When answering, think through the problem step by step: Step 1: Identify which policy documents are relevant to this question. Step 2: For each relevant policy, identify the specific sections that apply. Step 3: Extract the key requirements, thresholds, and conditions from those sections. Step 4: Determine whether the described scenario complies with or violates each requirement. Step 5: Identify what approvals or actions are needed for compliance. Step 6: Assess the overall compliance risk level (Low / Medium / High) based on the number and severity of compliance gaps. Step 7: Provide your final answer with citations to specific policy sections. Show your reasoning for each step before giving the final answer.

*Tokens: 3,255 · Response time: 7.87s · Output: 545 words · Highest coverage · Risk variance: Medium**

MODEL RESPONSE (GPT-4O), FULL STEPS EXECUTED, CONDENSED FOR BREVITY

Step 1: All three policies are relevant: POL-RW-2025, POL-DP-2025, POL-IS-2025. Step 2: - Remote Work Policy: §4.1 (general international approval), §4.2 (>30-day threshold), §4.3 (cross-policy compliance), §4.5 (extended

arrangements discouraged) - Data Privacy Policy: §5.1 (EEA transfer safeguards), §5.2 (approved transfer mechanisms), §5.4 (DPO approval for non-adequacy countries) - Information Security Policy: §4.1–4.2 (VPN requirements), §5.1 (full-disk encryption), §5.3 (personal device encryption verified by IT), §6 (BYOD) Step 3: Key thresholds, 30-day international limit (§4.2); EEA data cannot leave region without adequate safeguards (§5.1); DPO approval if processing PII from non-adequacy country (§5.4); BYOD device requires encryption, endpoint protection, remote wipe (§6.1–6.2). Step 4: Duration (45 days) violates §4.2 threshold → HR+Legal approval triggered. Personal device → BYOD registration required. Customer data → cross-border safeguard framework must be in place per §5.1. Step 5: Written manager approval + HR + Legal approval + BYOD registration with IT Security + DPO consultation on §5.4 applicability + encrypted cloud-only data access. Step 6: Risk Level: Medium, all requirements are actionable and achievable before departure. Step 7: Final answer with full citations: POL-RW-2025 §4.1, §4.2, §4.3, §4.5 | POL-DP-2025 §5.1, §5.2, §5.4 | POL-IS-2025 §4.1, §5.3, §6.1–6.2

Analytical Evaluation: CoT achieves the highest policy section coverage of all foundational techniques, uniquely surfacing §5.4 (DPO approval applicability) and §4.5 (extended arrangement discouragement) that no other technique retrieved. This validates Wei et al. (2022): step-by-step elicitation forces sequential document scanning that single-pass holistic generation misses. The 7-step scaffold's Step 3 ("extract thresholds and conditions") is specifically responsible for surfacing §5.4, as it directed the model to identify not just applicable sections but their triggering conditions, a qualitatively different retrieval task than section identification alone. However, CoT's most consequential finding is its anomalous risk classification: Step 6 outputs "Medium," where few-shot and analogical both output "High." The root cause is Step 6's framing, "assess risk based on number and severity of compliance gaps", which the model interpreted as residual risk after future compliance is achieved rather than the current non-compliance state. This single definitional ambiguity in one reasoning step creates a systematically different risk output. This is the most important diagnostic finding of the foundational phase and directly motivates the Self-Consistency advanced technique.

T4 · Step-Back Prompting, Categorical Abstraction Before Instantiation

SYSTEM PROMPT

Before answering the specific question, step back and first consider: 1. What are the general categories of compliance risk when an employee works internationally? (employment law, tax, data sovereignty, information security, benefits coverage) 2. What are the general principles governing personal device usage in a corporate environment? 3. What are the general principles governing cross-border handling of customer data? After identifying these broader principles, apply them to the specific scenario with citations.

Tokens: 2,893 · Response time: 6.38s · Output: 326 words · §4.4 and §4.5 uniquely surfaced

MODEL RESPONSE (GPT-4O), KEY CATEGORICAL OUTPUTS

Step-Back Principles Applied: Employment Law & Tax: 45-day Portugal arrangement may trigger local tax residency and employment law implications. §4.2 (HR+Legal approval) and §4.5 ("extended arrangements generally discouraged unless supported by business justification") directly address this risk category. Data Sovereignty: EEA-originated customer data requires adequate transfer safeguard framework per §5.1; employee must ensure all data remains within approved systems regardless of physical location (§5.3). Information Security: Personal device triggers BYOD policy obligations, device-level encryption, company-approved endpoint protection, remote wipe capability, and annual BYOD renewal (§6.1–6.5). Benefits Coverage: §4.4, the company does not guarantee health insurance or other employee benefits extend to international locations. Employee is responsible for verifying coverage prior to departure. Risk Level: High, four independent risk categories are simultaneously triggered. Employment law and tax domain, data sovereignty domain, information security domain, and benefits coverage domain are all active. Multi-domain simultaneous activation is a risk multiplier, not a risk average.

Analytical Evaluation: Step-back's categorical abstraction mechanism produced two unique policy discoveries: §4.4 (benefits coverage gap) and §4.5 (discouragement clause), both absent from all other foundational techniques. The mechanism is instructive, querying "what categories apply to international work generally?" generated a checklist that included "benefits coverage" as a domain, which then directed attention to §4.4. Query-direct approaches (zero-shot, CoT, few-shot) all failed to retrieve §4.4 because the test query did not mention benefits, the user's query anchored model attention to the domains the user considered relevant, not to the complete domain space a compliance review requires. Step-back overcomes query-anchored attention bias by instantiating a domain checklist independently of the query, then applying it. This is particularly valuable in multi-domain compliance scenarios where user queries are inherently incomplete, employees don't know what they don't know. The "risk multiplier vs. risk average" framing of multi-domain simultaneous activation was incorporated into the final prompt's risk level definition.

T5 · Analogical Prompting, Contrast-Based Risk Escalation Reasoning

SYSTEM PROMPT (LOW-RISK ANALOG PROVIDED AS REFERENCE)

Consider this analogous compliance scenario and how it was resolved: ANALOG: Employee works remotely from Canada for 20 days using a company-issued laptop with VPN. No customer data handled. Analysis: §4.1, 20 days within 30-day limit; manager approval only. §IS-4.1, company laptop + VPN satisfies network security. Data privacy: not triggered. Risk Level: Low, single policy domain, minimal approval chain. Now apply similar reasoning to the following scenario. Explicitly identify where the new scenario diverges from the analog and what additional policies those divergences trigger.

Tokens: 2,871 · Response time: 6.03s · Output: 292 words · Best explainability · Non-linear risk insight

MODEL RESPONSE (GPT-4O)

The Portugal scenario diverges from the Canada analog on three dimensions, each triggering additional compliance obligations: Divergence 1, Duration (45 vs. 20 days): Crosses 30-day threshold. §4.2 triggers HR and Legal department approval in addition to manager approval. Approval chain expands from 1 to 3 stakeholders. Divergence 2, Device (personal laptop vs. company laptop): §6.1–6.2 BYOD registration required. Device must have device-level encryption, company-approved endpoint protection software, and remote wipe capability enabled before any company data access. IT Security must verify and approve. Divergence 3, Data (customer data vs. no data): §5.1 EEA cross-border transfer safeguard framework activated. §5.3 cloud-only storage restriction applies, no local device storage regardless of encryption. §5.4 DPO consultation warranted to verify adequacy decision applicability to specific customer data flows. Compound Effect: Each divergence is independently actionable, but all three occur simultaneously, creating approval chain complexity (3 departments), data access restriction complexity (EEA framework + cloud-only), and device compliance complexity (full BYOD certification). The analog was Low risk with one approval. This scenario is High risk with three simultaneous compliance domains activated. Note: Risk escalation is non-linear. Three simultaneous Low-baseline divergences produce High, not Medium, because each independently triggers a separate policy domain with its own failure surface.

Analytical Evaluation: Analogical prompting produces the most pedagogically rigorous and explainable output of all foundational techniques. By making the compliance escalation mechanism explicit through contrast, three discrete divergences from a Low-risk baseline, the response teaches the reasoning process, not just the conclusion. The "non-linear risk escalation" insight (three simultaneous domain activations yield High, not the arithmetic average) is unique to this technique and was directly incorporated into the final system prompt's risk level definition. From a theoretical standpoint, analogical prompting works by providing a pre-validated inference template (the analog analysis) that the model uses to structure its reasoning about the target, reducing the variance in reasoning path selection that causes inconsistent risk classifications in other techniques. The primary limitation is analog dependency: the analog must be carefully chosen to share the relevant policy structure with the target scenario. A poorly chosen analog can mislead rather than guide.

T6 · Auto-CoT Prompting, Self-Generated Reasoning Framework

SYSTEM PROMPT

Before answering the question, do the following: 1. Generate the reasoning steps you would need to follow to thoroughly analyze this compliance question. List these steps explicitly. 2. Follow your own generated reasoning steps one by one to arrive at your answer. 3. Provide your conclusion with specific policy citations, required approvals, and a risk level assessment.

Tokens: 3,040 · Response time: 6.59s · Output: 468 words

SELF-GENERATED STEPS + CONCLUSION (GPT-4O)

Self-Generated Reasoning Steps: 1. Identify all policies applicable to international remote work + customer data handling + personal device use. 2. Review POL-RW-2025 for international duration thresholds and required approvals. 3. Review POL-DP-2025 for cross-border data handling requirements and applicable safeguards. 4. Review POL-IS-2025 for personal device and VPN network security requirements. 5. Determine which specific approvals are required based on duration and device type. 6. Assess combined compliance risk from all identified obligations. Conclusion: Required Approvals: Manager approval (§4.1) + HR and Legal approval for exceeding 30 days (§4.2) + IT Security BYOD registration (§6.1). Requirements: Device encryption, endpoint protection, remote wipe (§6.2); company-provided VPN updated to latest version (§4.1–4.2); customer data accessed through approved cloud systems only, encrypted if stored locally (§5.3). Risk Level: Moderate to High, personal device use and cross-border customer data handling create compounding security and privacy risk that must be addressed through concurrent approval and compliance processes.

Analytical Evaluation: Auto-CoT demonstrates that GPT-4o can generate functionally adequate reasoning scaffolds without expert-designed step sequences. The model's self-generated steps covered the same three policy domains as the manually crafted CoT scaffold, confirming that domain identification is within the model's zero-shot capability. However, the self-generated framework missed §5.4 (DPO approval applicability) and §4.4 (benefits coverage), both of which CoT's manually designed Step 3 ("extract thresholds and conditions") explicitly surfaced. This reveals a principled boundary: Auto-CoT generalizes well to high-salience reasoning patterns that are common in the training distribution but under-performs on domain-specific edge cases (DPO approval, benefits gaps) that require foreknowledge to include as scaffold targets. In production, Auto-CoT is preferable for scalability and generalization to novel query types; manually designed CoT is preferable for high-stakes compliance applications where known exhaustive coverage is a non-negotiable requirement. The §5.3 treatment is also imprecise ("encrypted if stored locally" rather than the policy's actual prohibition on local storage), a precision degradation later caught by Self-Criticism.

T7 · Generated Knowledge Prompting, Two-Step Compound Risk Modeling

TWO-STEP IMPLEMENTATION

STEP 1 PROMPT (Knowledge Generation): Generate relevant background knowledge about: (1) key compliance considerations for international remote work arrangements, (2) security implications of using personal devices for corporate data access, (3) data privacy considerations when handling customer data from foreign locations, (4) how these compliance domains interact and create compound risk. Do NOT answer the user's question yet. STEP 2 PROMPT (Grounded Response): Using the policy documents AND the generated background knowledge above, provide a comprehensive answer with specific citations, required approvals, and risk level assessment.

Step 1: 2,963 tokens · 5.39s | Step 2: 3,457 tokens · 5.75s | Total: 6,420 tokens · Compound risk framing unique

STEP 2 FINAL RESPONSE (GPT-4O)

Required Approvals: - Written manager approval (POL-RW-2025 §4.1) - HR and Legal department approval for exceeding 30 consecutive days (§4.2) - IT Security BYOD registration and device verification (POL-IS-2025 §6.1) Key Compliance Requirements: - POL-RW-2025 §4.3: Policy compliance is a mandatory condition of the arrangement, not a best-effort target - POL-DP-2025 §5.1: EEA personal data transfer requires adequate safeguard framework (SCCs, BCRs, or adequacy decision) before data access begins - POL-DP-2025 §5.3: Personal data accessed internationally must remain in approved cloud systems, not stored on local device - POL-IS-2025 §6.1–6.2: Device-level encryption, endpoint protection, remote wipe, all verified by IT Security prior to deployment - POL-IS-2025 §4.1–4.2: Company-provided VPN client required; must be updated to latest version before connection Risk Level: Moderate to High Reasoning: Three compliance domains activate simultaneously, extended international work triggers employment law and tax risk (HR/Legal domain); personal device triggers data breach risk (information security domain); customer data handling triggers regulatory risk (data privacy domain). Each domain is independently addressable, but their concurrent activation multiplies oversight complexity. The consequence of any single domain failure is elevated by the presence of the other two: a data breach on a personal device is materially worse when that device is in a foreign jurisdiction and processing customer PII subject to GDPR obligations.

Analytical Evaluation: Generated Knowledge is the only foundational technique to explicitly model compound risk, the emergent property by which three simultaneously active compliance domains interact to produce greater aggregate risk than their individual sums. The final sentence of the Step 2 response ("a data breach on a personal device is materially worse when in a foreign jurisdiction processing GDPR-subject PII") captures a non-additive risk interaction that none of the single-pass techniques articulated. This is theoretically grounded: compliance systems exhibit non-additive risk properties when multiple independent obligations share a common failure surface (in this case, the employee's personal laptop). The generated knowledge from Step 1 primed Step 2 to reason about interaction effects rather than independent domain checks. Cost: 6,420 tokens (2.3× the foundational average), justified only when compound risk modeling is the specific analytical objective, not as a general-purpose technique.

FIRST DRAFT SYSTEM PROMPT, SYNTHESIZED FROM FOUNDATIONAL TECHNIQUES

Integrates grounding strictness rules (T1), output behavioral schema from examples (T2), 7-step policy scan scaffold (T3), risk categorization framework (T4), compound risk framing (T7). Technique-specific mechanisms (analog, self-generation) are replaced by their transferable structural insights.

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// SAGE, INITIAL SYSTEM PROMPT (First Draft) You are a compliance policy assistant for TechNova Inc. Answer questions based ONLY on the following policy documents. Do not use external knowledge. Do not fabricate citations or policy sections. If the documents are insufficient to answer, state this explicitly. POLICY DOCUMENTS: [ALL_POLICIES] REASONING SEQUENCE (mandatory): Step 1: Identify all policy documents relevant to the question. Step 2: Locate specific applicable sections within each document. Step 3: Extract key requirements, thresholds, and conditions. Step 4: Assess compliance status of each requirement. Step 5: Enumerate all required approvals and corrective actions. Step 6: Assign a unified risk level (Low / Medium / High) with justification. RESPONSE FORMAT (always follow): Answer: [Comprehensive policy-grounded response] Citations: - [Document ID, Section, what it requires] Risk Level: [Low / Medium / High] Reasoning: [2–4 sentences connecting evidence to risk classification]
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3. Advanced System Prompt, Advanced Techniques

Advanced techniques were applied to stress-test, diagnose, and harden the first-draft system prompt. Each technique was executed programmatically against the same test query and evaluated for non-redundant improvements to the prompt's reliability, coverage, or precision. Every improvement listed below was incorporated into the final production system prompt.

A1 · Decomposition, 3-Stage Intent-to-Assessment Pipeline

PIPELINE ARCHITECTURE

Stage 1, Intent Classifier: Classify query as one of {risk_assessment | policy_question | out_of_scope}. Return category label only. No policy analysis. Stage 2, Policy Analyst: For each applicable policy section, extract the specific requirement and classify as: COMPLIES / VIOLATES / REQUIRES ACTION. Do not generate a final answer, produce only the structured compliance audit. Stage 3, Risk Assessor: Synthesize Stage 2 output into final structured response (Answer / Citations / Risk Level / Reasoning).

Stage 1: 143 tokens · 0.36s | Stage 2: 3,012 tokens · 5.58s | Stage 3: 1,098 tokens · 3.64s | Total: 4,253 tokens

STAGE 1 OUTPUT

risk_assessment

STAGE 2 OUTPUT, COMPLIANCE STATUS TAXONOMY

POL-RW-2025 §4, REQUIRES ACTION: 45-day duration exceeds 30-day international limit; HR + Legal approval mandatory before departure. POL-RW-2025 §4.3, REQUIRES ACTION: Compliance with POL-DP-2025 and POL-IS-2025 must be verified and documented before arrangement begins. POL-RW-2025 §5.2, REQUIRES ACTION: Personal device use requires prior IT Security team approval separate from manager approval. POL-DP-2025 §5, REQUIRES ACTION: EEA cross-border data transfer safeguard framework must be established before customer data is accessed from Portugal. POL-IS-2025 §4.1, COMPLIES: Employee intends to use company VPN. (Only pre-compliant element identified.) POL-IS-2025 §6, REQUIRES ACTION: BYOD registration required; device must meet encryption, endpoint protection, remote wipe standards. POL-IS-2025 §5, REQUIRES ACTION: Personal device encryption must be independently verified by IT Security before data access.

STAGE 3 FINAL RESPONSE

Answer: Employee must complete six compliance actions before the arrangement begins: manager approval, HR approval, Legal department approval, IT Security BYOD registration with device verification, EEA data transfer safeguard establishment, and benefits coverage verification. VPN compliance is the only pre-satisfied requirement. Risk Level: High Reasoning: Six of seven policy requirements are in REQUIRES ACTION status prior to departure. The only compliant element is VPN intent. Multi-department approval chain, data privacy safeguard framework, and full BYOD certification must all be completed concurrently, creating execution risk through coordination dependency.

Analytical Evaluation: Decomposition's primary contribution is the COMPLIES / VIOLATES / REQUIRES ACTION compliance status taxonomy, a structured audit output not produced by any holistic foundational technique. Stage 2's finding that only VPN intent was pre-compliant provided a precise, quotable risk justification: "6 of 7 requirements in REQUIRES ACTION status" is a materially more actionable risk characterization than "High risk due to multiple obligations." The taxonomy enables SAGE to communicate not just risk level but compliance gap granularity, transforming a risk classification into an actionable remediation checklist. Additionally, Stage 1's 143-token intent classification (0.36s) demonstrates that lightweight query routing can prevent the full reasoning pipeline from being invoked on out-of-scope queries, meaningfully reducing average computational cost per request in production. Both contributions, compliance status taxonomy and intent routing, were incorporated directly into the final prompt architecture.

A2 · Ensembling, 3 Perspectival Variants with Conservative Aggregation

VARIANT DESIGN

Variant A, Legal & Regulatory: Analyze the scenario from a legal and regulatory implications perspective. Emphasize statutory obligations, potential legal exposure, and regulatory authority requirements. Variant B, Information Security: Analyze from an information security and data protection perspective. Emphasize access controls, encryption requirements, device management, and breach risk. Variant C, Operational & HR: Analyze from an operational and HR perspective. Emphasize approval processes, employee obligations, HR liability, and operational feasibility. Aggregation Instruction: Compare all three analyses. Resolve contradictions by adopting the most conservative (risk-averse) interpretation. Identify any policy coverage present in one variant but absent from others.

Variant A: 2,839 tokens | Variant B: 2,669 tokens | Variant C: 2,887 tokens | Aggregation: 2,281 tokens | Total: 10,676 tokens

NON-REDUNDANT UNIQUE FINDING PER VARIANT

Variant A (Legal), Unique finding: GDPR applicability must be stated explicitly. Portugal is an EEA member state; §5.1 requires an adequate safeguard framework (Standard Contractual Clauses, Binding Corporate Rules, or adequacy decision) to be in place before any personal data is processed from Portugal. Customer data originating from EEA subjects triggers GDPR Article 46 transfer mechanism requirements even though Portugal itself has adequacy within the EEA. Variant B (Security), Unique finding: §5.4 (Data Privacy Policy), "processing customer PII from a country without an adequacy decision requires prior approval from the Data Protection Officer." Portugal has EEA adequacy, but the DPO must verify that the specific customer data flows in question are covered. The DPO consultation step is warranted regardless of Portugal's adequacy status. Variant C (HR/Ops), Unique finding: §4.4 creates a documented employer liability exposure, not merely a procedural item. If an employee working internationally under an approved arrangement experiences a medical emergency and discovers their health insurance does not extend to Portugal, the absence of a documented benefits verification step creates an organizational liability. §4.5 additionally requires a documented business justification for any international arrangement, this is an HR documentation obligation separate from the approval chain. Aggregated Risk Level: High (maximum across all three variants; conservative synthesis applied)

Analytical Evaluation: Ensembling exposed three non-overlapping coverage gaps, one per variant, that the initial single-framing prompt missed. The mechanism is theoretically expected: each framing activates a different priority ordering over the same document corpus, retrieving domain-specific content that other framings deprioritize during single-pass generation. Variant C's HR liability framing of §4.4 was the most consequential new finding, it transformed a procedural item ("employee must verify benefits") into a documented employer liability concern, which is a qualitatively different risk characterization with implications for how SAGE should present the requirement to users. The aggregation step's conservative synthesis rule (adopt the highest risk among variants; surface all unique findings) is critical: without it, averaging would systematically understate compound risk. The primary cost is token volume (10,676 tokens, the highest of all techniques), appropriate for prompt QA during development, not for per-query production execution.

A3 · Self-Consistency, Variance Detection via Independent Sampling

SETUP: INITIAL SYSTEM PROMPT, N=3 INDEPENDENT RUNS, TEMPERATURE=0.5

Run 1: 2,919 tokens | Run 2: 2,906 tokens | Run 3: 2,920 tokens | Total: 8,745 tokens

RISK LEVEL OUTPUTS ACROSS 3 RUNS

Run 1 → Risk Level: HIGH Justification: Multiple simultaneous compliance gaps across all three policies, BYOD certification, data privacy safeguards, and multi-department approval requirements compound the risk. Current state

is non-compliant across six of seven identified requirements. Run 2 → Risk Level: MEDIUM Justification: Employee can potentially comply with all requirements through proper process execution. Risk is elevated but manageable, all obligations have defined resolution paths. Risk assessment reflects achievable post-compliance state. Run 3 → Risk Level: HIGH Justification: Multiple approvals required across three departments, data security obligations, and BYOD compliance requirements collectively constitute high compliance risk. Concurrent multi-domain obligation creates coordination execution risk. Variance Detection: 2/3 runs → High | 1/3 runs → Medium Majority vote → High (correct output) Root cause of variance: Temporal framing ambiguity in risk definition.

Analytical Evaluation: Self-consistency's most important contribution is not the majority vote result, which correctly selected High, but the diagnostic finding it enabled. The variance between Runs 1/3 (High) and Run 2 (Medium) is not stochastic noise; it is a deterministic semantic ambiguity in the prompt's risk definition. Run 2 interpreted "assess risk level" as residual risk assuming future compliance is achieved; Runs 1 and 3 interpreted it as current non-compliance state before corrective actions. Both interpretations are defensible readings of the prompt, the ambiguity is in the prompt language, not in the model's reasoning. This transforms Self-Consistency from a voting mechanism into a diagnostic instrument: it reveals that the prompt has an ambiguous risk definition that will systematically produce different outputs depending on the reasoning path selected. The fix is definitional precision, not repeated sampling: adding "based on current non-compliance state, before corrective actions are taken" to the risk level instruction eliminates the variance entirely. This is the methodologically correct use of Self-Consistency, run during prompt development, identify ambiguities, fix the prompt, and eliminate the need for sampling at inference time.

A4 · Universal Self-Consistency, LLM-as-Evaluator Cross-Run Synthesis

EVALUATOR PROMPT

You are a compliance quality reviewer. Three responses were generated for the same query using the same system prompt. Compare all three for: (1) consistency, where do they agree and disagree? (2) completeness: Are any applicable policy sections present in some responses but absent in others? (3) accuracy, are the cited sections correctly interpreted? Resolve all inconsistencies by adopting the most defensible interpretation grounded in the current non-compliance state, not potential future compliance. Produce a single definitive synthesized response.

Evaluator: 1,800 tokens · 5.55s | Cumulative with SC runs: 10,545 tokens

EVALUATOR ANALYSIS OUTPUT

Consistency Analysis: CONSISTENT across all three runs: Manager + HR + Legal approval requirement (§4.2); BYOD registration and device requirements (§6.1–6.2); data privacy compliance obligation (§4.3, §5.3). INCONSISTENT: Risk level (2× High, 1× Medium), attributed to temporal framing ambiguity in Step 6 of the reasoning scaffold. Resolution: assess risk based on the current non-compliance state. → Risk Level: High confirmed. COVERAGE GAP IDENTIFIED: VPN compliance status (§4.1, POL-IS-2025) was explicitly cited and assessed in only Run 3. Absent from Runs 1 and 2 despite being directly relevant. VPN is the only pre-compliant element; its absence from most responses creates an incomplete and misleadingly negative picture. Definitive Synthesized Response: [Full structured response incorporating all consistent findings, resolved risk classification, and added VPN compliance status] Final Risk Level: High, anchored to current non-compliance state (6/7 requirements REQUIRES ACTION).

Analytical Evaluation: Universal Self-Consistency contributed two improvements beyond what SC alone provided. First, the evaluator's completeness audit identified a cross-run coverage gap, VPN compliance absent from 2/3 runs, that majority voting cannot detect. Voting aggregates answers but does not audit their completeness relative to a ground truth policy set. Second, the evaluator formalized the risk level resolution criterion ("current non-compliance state") that SC had diagnosed but not resolved. The LLM-as-evaluator

pattern here instantiates a meta-reasoning level: one model completion evaluating the outputs of three others, operating at the abstraction level of a compliance review supervisor. This is the operational principle underlying Constitutional AI critique phases and multi-agent review pipelines. The practical prompt engineering implication: Universal SC is most valuable when run after Self-Consistency during development, as a synthesis step that transforms the diagnostic findings into concrete prompt language improvements.

A5 · Self-Criticism, Generate → Critique → Revise (3-Step Validation Pipeline)

CRITIQUE EVALUATION CRITERIA

Evaluate the initial system prompt response on five dimensions: (1) GROUNDEDNESS: Is every claim directly supported by specific policy text? Identify any claims not grounded in a cited section. (2) CITATION ACCURACY: Do the cited section numbers say what the response claims they say? Check each citation individually. (3) COMPLETENESS: Are there applicable policy sections that should have been included but were not? Check all three documents systematically. (4) RISK LEVEL JUSTIFICATION: Is the risk classification proportional to the evidence presented? Is it too high, too low, or appropriately calibrated? (5) AMBIGUITY HANDLING: Where policy language is ambiguous or conditions are uncertain, does the response flag the ambiguity rather than asserting a specific interpretation?

Critique: 3,319 tokens · 3.27s | Revision: 3,694 tokens · 5.28s | Total: 9,935 tokens

CRITIQUE OUTPUT, GPT-4O SELF-AUDIT FINDINGS

Groundedness: No issues identified. All claims in the response accurately reflect the text of the policy documents. No extrapolation or fabrication detected. Citation Accuracy: No errors. All cited section numbers correctly map to the requirements attributed to them. Missing Coverage, Three gaps identified: GAP 1: POL-RW-2025 §4.4, Response does not mention that the company does not guarantee employee benefits, including health insurance, extend to international locations. The employee is responsible for verifying coverage before departure. This is a material omission for an employee relying on SAGE for pre-departure compliance guidance. GAP 2: POL-DP-2025 §5.1, Response cited §5.3 (encrypted storage requirement) but did not cite §5.1 (the EEA cross-border adequate safeguard framework prerequisite). §5.3 is contingent on §5.1 compliance; omitting §5.1 creates an incomplete picture of the data privacy obligation hierarchy. GAP 3: POL-IS-2025 §5.3, PRECISION ERROR: Response states "not stored on local devices unless encrypted." The actual policy states, "employees must not store company data locally on the device. All data must be accessed through approved cloud-based applications and saved to company-managed storage." This is not a conditional permission for encrypted local storage; it is an unconditional prohibition on local storage. The response softened a hard prohibition into a conditional, which could lead an employee to incorrectly conclude that encrypted local storage is policy-compliant. Risk Level: Appropriately assessed as High. Justification is sound. Ambiguity: §5.4 DPO applicability (Portugal has EEA adequacy; question is whether specific data flows require DPO sign-off) should be flagged as requiring DPO consultation rather than assumed compliant.

REVISED RESPONSE ADDITIONS (STEP 3)

ADDED, POL-RW-2025 §4.4: The company does not guarantee health insurance or other employee benefits extend to international locations. Employee must verify coverage before departure. Failure to do so creates both personal risk and potential organizational liability. CORRECTED, POL-DP-2025 §5.1: EEA cross-border data transfer adequate safeguard framework (Standard Contractual Clauses, Binding Corporate Rules, or adequacy decision) must be confirmed as in place before any customer data is accessed from Portugal. This is a prerequisite condition for §5.3, not a parallel requirement. CORRECTED, POL-IS-2025 §5.3: Company data must NOT be stored locally on the personal device under any circumstances. All data must be accessed through approved cloud-based applications and saved to company-managed cloud storage. Encryption of locally stored data does NOT satisfy this requirement; local storage itself is prohibited. FLAGGED, POL-DP-2025 §5.4: DPO consultation is recommended to verify that the specific customer data flows involved are covered by Portugal's EEA adequacy status for the data types in question.

Analytical Evaluation: Self-criticism's most significant finding was Gap 3: a precision error in the initial response's treatment of §5.3. The response stated "not stored locally unless encrypted", a conditional accommodating local storage if encrypted, when the actual policy is an unconditional prohibition: "must not store company data locally on the device." This is not a hallucination (the model did not fabricate a policy that doesn't exist) and not a citation error (the section was correctly identified). It is a systematic precision degradation: the model softened a hard prohibition into an accommodating condition, likely because this reflects common practical behavior that the model generalizes from training data. In a compliance context, this difference is material and dangerous; an employee who acts on SAGE's "encrypted local storage is acceptable" interpretation would be in clear policy violation. Standard hallucination detection would not flag this, because no content was fabricated. Self-criticism with explicit groundedness criteria, requiring that every claim be traced to the specific policy language, not just the section, is the appropriate and necessary mechanism for detecting sub-hallucination precision degradation. This finding alone justifies self-criticism as a mandatory component of any compliance AI validation pipeline.

FINAL PRODUCTION-READY SYSTEM PROMPT, INCORPORATING ALL 12 TECHNIQUE IMPROVEMENTS

A1: intent routing + compliance status taxonomy | A2: §4.4 HR liability + DPO trigger | A3: current-state risk anchoring | A4: VPN mandatory coverage | A5: §5.1 prerequisite, §5.3 corrected prohibition, §4.4 mandatory, DPO flagged

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// SAGE, FINAL ADVANCED SYSTEM PROMPT You are a compliance policy assistant for TechNova Inc. Answer questions based ONLY on the following policy documents. Do not use external knowledge. Do not fabricate sections, citations, or requirements. If documents are insufficient to answer, state this explicitly. POLICY DOCUMENTS: [ALL_POLICIES] INTENT ROUTING (classify before reasoning): risk_assessment → employee describes a scenario for compliance evaluation policy_question → employee asks about a specific policy or procedure out_of_scope → decline politely; do not apply policy reasoning MANDATORY REASONING SEQUENCE: Step 1: Identify all relevant policy documents. Step 2: Locate applicable sections and tag each with compliance status: COMPLIES | VIOLATES | REQUIRES ACTION Step 3: Explicitly check these mandatory items in every risk assessment response: · POL-IS-2025 §4.1, VPN compliance status (state COMPLIES or REQUIRES ACTION) · POL-RW-2025 §4.4, Benefits and health insurance coverage gap for international arrangements · POL-DP-2025 §5.1, EEA cross-border transfer safeguard framework (prerequisite to §5.3) · POL-IS-2025 §5.3, Local storage PROHIBITION: data must not be stored locally under ANY circumstance; cloud-only access required; encrypted local storage does NOT satisfy this requirement · POL-DP-2025 §5.4, DPO consultation applicability for customer PII data flows Step 4: Enumerate all required approvals with responsible stakeholders. Step 5: Assign risk level based on CURRENT NON-COMPLIANCE STATE before corrective actions: Low → 1 policy domain, clear single-step resolution Medium → 2 domains, defined multi-step resolution path High → 3+ domains OR multi-department approval chain OR data privacy + security combined RESPONSE FORMAT: Answer: [Policy-grounded response; all applicable requirements covered] Citations: - [Document ID, §Section, Requirement, Compliance Status] Risk Level: [Low / Medium / High], [current-state justification] Reasoning: [2–4 sentences; cite specific sections driving the classification]
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4. Reflection on Prompt Effectiveness

4.1 Quantitative Comparison Across All Twelve Techniques

Technique	Type	Tokens	Risk Output	Unique Contribution	Critical Limitation
T1: Zero-Shot	Foundational	2,721	Fragmented	Accurate policy identification; functional baseline	No unified risk; §4.4, §5.1 absent
T2: Few-Shot	Foundational	2,807	High ✓	Best format fidelity; correct unified risk classification	Coverage bounded by example domain emphasis
T3: Chain-of-Thought	Foundational	3,255	Medium*	Highest section coverage; §5.4 DPO uniquely surfaced	Risk ambiguity (temporal framing)
T4: Step-Back	Foundational	2,893	High ✓	§4.4 benefits gap + §4.5 via categorical scan	Quality depends on the abstraction category design
T5: Analogical	Foundational	2,871	High ✓	Non-linear risk escalation insight; best explainability	Quality constrained by analog selection
T6: Auto-CoT	Foundational	3,040	Mod–High	Autonomous scaffold; scalable to new query types	Misses domain-specific edge cases (§5.4, §4.4)
T7: Gen. Knowledge	Foundational	6,420	Mod–High	Compound risk modeling; emergent interaction framing	2.3× token cost; slowest foundational technique
A1: Decomposition	Advanced	4,253	High ✓	Compliance status taxonomy; intent routing efficiency	3 API calls; production latency impact
A2: Ensembling	Advanced	10,676	High ✓	§4.4 HR liability framing; DPO trigger; max coverage	4 API calls; highest total token cost
A3: Self-Consistency	Advanced	8,745	2×H, 1×M	Risk definition ambiguity was diagnosed as the root cause	Diagnostic only; not a per-query solution
A4: Universal SC	Advanced	10,545	High ✓	VPN coverage gap; risk criterion formalized	Development-phase use; not production inference

A5: Self-Criticism	Advanced	9,935	High ✓	§5.3 precision error; 3 missing sections; no hallucination confirmed	Critique quality depends on the evaluator's criteria

4.2 Most Effective Techniques, Ranked Analysis

Rank 1: Self-Criticism, Self-criticism is the highest-impact technique in this study because it detected a class of error that no other technique can identify: sub-hallucination precision degradation. The discovery that the initial response had systematically softened §5.3's unconditional local storage prohibition into a conditional permission ("encrypted if stored locally") is a finding of material compliance consequence. An employee acting on this softened version would be in policy violation while believing they were compliant. The model did not fabricate content; it accurately retrieved the section but imprecisely paraphrased the prohibition as a conditional. This error pattern is systematic, not random: LLMs trained on vast corpora of human-written content will generalize toward accommodating phrasings because hard prohibitions are less common in training data than conditional permissions. Self-criticism with precisely specified evaluation criteria (groundedness, citation accuracy, completeness, ambiguity handling) is the appropriate architectural mechanism for catching this class of error before it reaches users. Its ranking at the top reflects not token efficiency (9,935 tokens) but irreplaceability: no other technique in this study caught the §5.3 precision error.

Rank 2: Few-Shot Prompting, Few-shot is the highest-value foundational technique by the ratio of quality improvement to implementation effort. Two labeled examples were sufficient to produce the only foundational response with a correct unified risk classification, proper structured output format, and actionable reasoning, improvements that explicit structural instructions in zero-shot failed to achieve. The mechanism is theoretically well-understood: in-context behavioral demonstrations shift the model's output distribution more reliably than declarative format instructions for structured generation tasks, because examples communicate both the surface structure and the inferential logic simultaneously. The risk classification accuracy, High, with correct compound justification, is particularly notable: the model inferred from the examples' risk reasoning that multi-policy multi-domain violations warrant elevated classification, without being explicitly instructed on this decision boundary. Few-shot's limitation is coverage: the examples' implicit domain emphasis bounded what the model retrieved, missing §4.4 and §4.5. This limitation is correctable by improving the example design; it is not intrinsic to the technique.

Rank 3: Self-Consistency, Self-consistency ranks third not because of its output quality (it does not produce the final response) but because of the diagnostic finding it generated: the identification of a prompt-level semantic ambiguity as the root cause of risk classification variance. The discovery that "assess risk level" is interpretable as either current-state or future-state risk, and that different reasoning paths select different interpretations deterministically, is a finding that fundamentally changes how the prompt should be written. The fix ("based on current non-compliance state, before corrective actions are taken") is precise and verifiable. Without self-

consistency, this ambiguity would have been invisible until it caused a disagreement between two different users receiving different risk assessments for the same scenario. Self-consistency transforms that future incident into a development-time prompt quality finding.

Rank 4: Chain-of-Thought, CoT ranks fourth because of its unique coverage depth, despite its risk classification anomaly. The 7-step scaffold is the only mechanism in this study that reliably surfaces §5.4 (DPO approval applicability), a domain-specific edge case that requires directed attention ("extract thresholds and conditions") rather than general section identification. The insight that manually designed CoT steps outperform auto-generated steps for domain-specific edge cases is an important system design finding: Auto-CoT is preferable for generalizability, but expert-designed CoT scaffolds are preferable for exhaustive coverage in high-stakes compliance applications. CoT's risk classification anomaly, Medium rather than High, was fully diagnosed by Self-Consistency and resolved through definitional precision, so it does not diminish CoT's net contribution to the final prompt.

4.3 How Few-Shot and Chain-of-Thought Shaped Response Quality

Few-shot and chain-of-thought address orthogonal quality dimensions, a finding with direct architectural implications for production compliance prompt design. This is not a rhetorical claim but a finding grounded in the measurable output differences across conditions.

Few-shot's primary effect is on output format and risk classification accuracy, not on policy coverage. Comparing zero-shot (no examples) to few-shot (two examples), the format transformation is dramatic: from fragmented domain-specific sub-risks to a unified structured response with consistent section citations and a single justified risk level. The content improvement is more modest; few-shot correctly identifies all three policy documents and the 30-day threshold, as did zero-shot. The risk classification accuracy improvement is significant: zero-shot produced a fragmented "Moderate" assessment across three sub-domains; few-shot produced a unified "High" with a compound justification that correctly captures the multi-domain interaction. This classification improvement derives from the examples' implicit demonstration of the risk decision boundary; two labeled examples that showed Low (single domain, clear resolution) and Medium (procedural violation, remediable) were sufficient for the model to infer that multi-policy multi-domain violations warranted High. The few-shot examples functioned as a risk calibration instrument, not merely a format template.

Chain-of-thought's primary effect is on policy coverage completeness, not on output format or risk accuracy. Comparing zero-shot to CoT, the format change is modest (both produce reasonably structured output). The coverage change is substantial: CoT uniquely surfaced §5.4 (DPO approval) and §4.5 (extended arrangement discouragement), both of which have zero-shot and few-shot misses. The mechanism is the 7-step mandatory scan, specifically Step 3 ("extract key thresholds and conditions") and Step 2 ("identify the specific sections that apply within each document"), which directed the model to scan sections it would not have attended to in a single-pass generation. CoT's cost is the risk classification anomaly: its Step 6 framing ("assess risk based

on number and severity of compliance gaps") was interpreted as residual post-compliance risk, yielding Medium. This reveals that CoT's sequential reasoning is vulnerable to step-level framing effects; the interpretation of a single step's instruction can determine the output of subsequent steps. The practical implication: CoT steps must be phrased with the same precision as the final system prompt they are designed to produce.

The production design conclusion is that few-shot and CoT should be combined rather than selected between. Few-shot establishes the behavioral output schema and calibrates the risk decision boundary through example demonstration. CoT's sequential scanning scaffold ensures exhaustive policy coverage that single-pass generation misses. The two techniques are not redundant; they address format and completeness, respectively, and their combination in the final prompt architecture produces a response that is simultaneously well-structured, accurately classified, and exhaustively policy-grounded.

4.4 Analytical Evaluation of Most Effective Techniques, Theoretical Grounding

The findings of this study are interpretable within established theoretical frameworks for LLM prompting. Wei et al. (2022)'s demonstration that chain-of-thought elicitation produces better performance on complex multi-step tasks is confirmed in the policy coverage dimension: the sequential step structure forces the model through a reasoning path that single-pass generation would short-circuit. The coverage gap between CoT (which surfaces §5.4, §4.5) and all other single-pass techniques directly reflects Wei et al.'s finding that elicited intermediate reasoning steps enable the retrieval of information that holistic generation misses.

The few-shot behavioral schema transfer result aligns with the in-context learning literature (Brown et al., 2020): GPT-scale models can infer task format, domain conventions, and decision boundaries from a small number of labeled examples without gradient updates. The two examples provided were sufficient to shift the output distribution toward the target compliance response schema, confirming that behavioral demonstration is the appropriate mechanism for format specification, not explicit structural instruction.

The self-criticism precision degradation finding, where the model softened a prohibition into a conditional, is consistent with the hallucination mitigation literature's growing recognition that LLMs exhibit a spectrum of accuracy errors from full fabrication to subtle paraphrase distortion. Maynez et al. (2020) and subsequent work distinguish between faithfulness failures (fabrication) and hallucination in summarization tasks; the §5.3 precision error is analogous to a faithfulness failure in the paraphrase direction: the model accurately retrieved the section but distorted its semantic content toward a more accommodating form. Self-criticism with specific semantic accuracy criteria is the appropriate detection mechanism because it evaluates not just grounding (is the section cited?) but fidelity (does the paraphrase accurately preserve the policy's semantic content and modality?).

The self-consistency variance finding is interpretable as a prompt ambiguity detection mechanism. Wang et al. (2022)'s self-consistency framework was developed as an aggregation method for improving accuracy; this study demonstrates an additional use: when self-consistency produces high variance across runs at low temperature, the variance itself is diagnostic of prompt ambiguity rather than model uncertainty. The Medium/High risk level variance in this study was caused by an ambiguous instruction ("assess risk level") that admitted two valid interpretations, not by stochastic noise in the model's generation. Treating variance as a prompt defect signal rather than a sampling artifact is a methodologically important reframing with practical implications for prompt engineering workflows.

4.5 Technique Pairing and Design Principles

The twelve techniques in this study fall into two functional categories with distinct roles in the prompt engineering workflow. **Generative techniques** (T1–T7, A1–A2) produce responses or knowledge artifacts that contribute structural, coverage, or framing improvements to the system prompt. **Diagnostic techniques** (A3–A5) evaluate existing artifacts to surface defects, ambiguities, and coverage gaps that generative techniques cannot self-detect.

Within the generative category, the most productive pairings are **Few-Shot + CoT** (format schema + coverage depth, addressing orthogonal dimensions), **Step-Back + Generated Knowledge** (categorical coverage + compound interaction modeling), and **Analogical + Step-Back** (escalation explanation + domain breadth). These pairings are complementary rather than redundant; each addresses a dimension the other does not cover. Within the diagnostic category, the most productive sequential application is **Self-Consistency → Universal SC → Self-Criticism**: SC identifies variance and its root cause; Universal SC synthesizes the SC runs and formalizes the correction; Self-Criticism audits the corrected prompt output for precision and coverage completeness. Each diagnostic technique operates at a different abstraction level: SC at the prompt semantic level, Universal SC at the cross-run synthesis level, and Self-Criticism at the output content fidelity level.

The key design principle for compliance AI prompt engineering emerging from this study: diagnostic techniques should be applied sequentially after the generative techniques have been combined, and their findings should produce concrete prompt language changes rather than inference-time workarounds. Self-Consistency diagnosed the risk definition ambiguity; the correct response was to rewrite the risk level instruction with temporal anchoring, not to add sampling at inference time. Self-Criticism identified the §5.3 precision gap; the correct response was to add an explicit mandatory checklist item with the precise prohibition language, not to add a post-generation audit step to every production query. The goal of the advanced technique phase is to eliminate the defects from the prompt itself, not to build compensatory mechanisms around them.

4.6 Token Economics and Production Deployment Considerations

The foundational techniques consumed an average of 3,152 tokens per technique (range: 2,721–6,420). The advanced techniques consumed an average of 8,791 tokens per technique (range: 4,253–10,676), reflecting their multi-call architectures. The final production system prompt, when executed in a single-call architecture against the TechNova policy corpus, consumes approximately 2,900 tokens per query, comparable to the foundational baseline. This is the critical deployment insight: the advanced techniques are development-time investments, not per-query costs. Self-Consistency (8,745 tokens for 3 runs) was used to diagnose the risk definition ambiguity and resulted in a two-word addition to the prompt ("current non-compliance state") that eliminates the need for multi-run sampling in production. Ensembling (10,676 tokens for 4 calls) was used to surface §4.4, §5.1, and DPO coverage gaps and resulted in mandatory checklist additions that are verified in every single-call production query. The return on the development-time token investment is a production prompt that is simultaneously more accurate, more complete, and more computationally efficient than any of the techniques that produced it.

5. Conclusion

This assignment demonstrates that systematic multi-technique prompt engineering produces qualitatively superior results that cannot be achieved through expert judgment or any individual technique in isolation. The evolution from zero-shot baseline to final production system prompt incorporated twelve discrete improvements, each attributable to a specific technique and each non-redundant with the others: zero-shot provided the policy identification baseline; few-shot calibrated the risk classification decision boundary and output schema; chain-of-thought ensured exhaustive sequential policy scanning; step-back eliminated query-anchored attention bias; analogical reasoning produced the non-linear risk escalation insight; generated knowledge introduced compound risk modeling; decomposition contributed the compliance status taxonomy and intent routing; ensembling surfaced three non-overlapping domain-specific coverage gaps; self-consistency diagnosed a prompt-level semantic ambiguity as the root cause of risk classification variance; universal self-consistency resolved the ambiguity and identified a cross-run coverage gap; and self-criticism detected a sub-hallucination precision degradation that no other technique could identify.

The final SAGE system prompt is production-ready for the TechNova policy corpus: it is grounded (zero fabricated claims confirmed), precise (§5.3 prohibition correctly stated), complete (all known applicable sections included via mandatory checklist), and auditable (every response carries section-level citations with compliance status tags). The prompt engineering methodology documented here, systematic technique application, cross-technique comparison on a consistent test query, and diagnostic technique application to refine generative outputs, is generalizable to enterprise compliance AI applications beyond this specific domain.

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